Calibration of Merton and Kou Jump-Diffusion Models to QQQ

Stochastic Processes — Final Group Project — Workstream 6

Group 3

Workstream 6: Calibration Engineer

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Data and Methodology

We calibrate the Merton (1976) and Kou (2002) jump-diffusion models to a snapshot of QQQ option prices taken on 29 May 2026, with spot price $S_0 = \$735.60$ and a risk-free rate of $r = 5.00\%$. QQQ ETF options are American-style; we approximate them as European because the calibration uses only out-of-the-money quotes and excludes deep in-the-money contracts, where the early-exercise premium is largest. This approximation is documented as a limitation in the closing subsection. The raw option chain was retrieved through `yfinance` for eight maturities spread across the term structure (one week, two weeks, one month, two months, three months, seven months, ten months, thirteen months); 2,480 quotes survived the basic data-quality filters of `src/data_pipeline.py` (positive bid-or-last price, non-stale, and IV plausibly below 300%).

For calibration we restrict to a clean near-ATM subset by applying four additional filters: time to expiry $T \geq 0.05$ years (drops the two shortest weekly maturities), out-of-the-money quotes only (puts with $K < S_0$ and calls with $K > S_0$), a moneyness band $K/S_0 \in [0.85, 1.15]$, and a Brent-recomputed implied volatility in $[0.10, 0.60]$. After cleaning, $M = 424$ quotes across six maturities remained, ranging from $T = 0.077$ y (one month) to $T = 1.052$ y (thirteen months).

For each candidate parameter vector θ, model prices are computed via the COS method of Fang and Oosterlee (2008) with $N_{\mathrm{COS}} = 512$ expansion terms and truncation parameter $L = 10$. Each model price is inverted back to its Black–Scholes implied volatility via Brent's method, and we minimise the vol-space root mean squared error

$$\mathrm{RMSE}(\theta) \;=\; \sqrt{\frac{1}{M}\sum_{i=1}^{M}\big(\sigma_i^{\mathrm{IV,model}}(\theta) - \sigma_i^{\mathrm{IV,market}}\big)^2},$$

where M is the number of cleaned quotes (here $M = 424$) and N_{COS} controls the COS Fourier-series accuracy.

Optimisation runs in two stages. A `differential_evolution` global search (population size 12, 40 generations, evaluated on a stratified sub-sample of 96 quotes for speed) locates the right region in parameter space; an `L-BFGS-B` local refinement then polishes the optimum on the full 424 quotes. All parameter bounds and optimiser settings are read from `.env` through the `CalibrationConfig` dataclass.

We acknowledge two limitations on the data side. The snapshot was taken outside NYSE trading hours, so 98% of cleaned quotes use last-trade price as a fallback for mid-price. We mitigate this by recomputing all implied volatilities ourselves via Brent rather than trusting vendor-reported IVs, which are unreliable for stale or zero-bid quotes. QQQ also pays a quarterly dividend (˜0.6% annualised); we do not subtract a dividend yield q from the discount rate. This biases the implied σ marginally upward for longer-dated options; a production calibration should be performed on forward prices.

Merton Calibration

The calibrated Merton parameters are summarised in the table below. The achieved vol-space RMSE is 0.01400, an average pricing error of 1.40 percentage points in implied volatility across the 424-quote surface.

Table 1: Calibrated Merton parameters (QQQ, 2026-05-29).

Parameter	Value	Interpretation
σ	0.1858	continuous diffusion volatility
λ	0.1633	risk-neutral jump intensity (jumps per year)
μ_J	-0.4954	mean log-jump size (at lower bound -0.50)
σ_J	0.3339	log-jump size standard deviation
κ	-0.3558	mean percentage jump $\mathbb{E}[e^J - 1]$
RMSE	0.01400	vol-space RMSE on 424 quotes

Two features deserve immediate comment. First, the diffusion volatility $\sigma \approx 19\%$ is consistent with the realised volatility of QQQ over the past three years and with the at-the-money level of the IV surface. Second, the optimum lands on the lower bound of the project's specified μ_J range: even when the bound is widened to -0.50, the optimiser still attaches itself to it, implying a calibrated model in which crashes are very rare ($\lambda = 0.16$, i.e., roughly one event every six years) but catastrophic in magnitude ($\kappa \approx -36\%$ on average). This boundary behaviour is the empirical signature of the identification problem analysed in the next subsection.

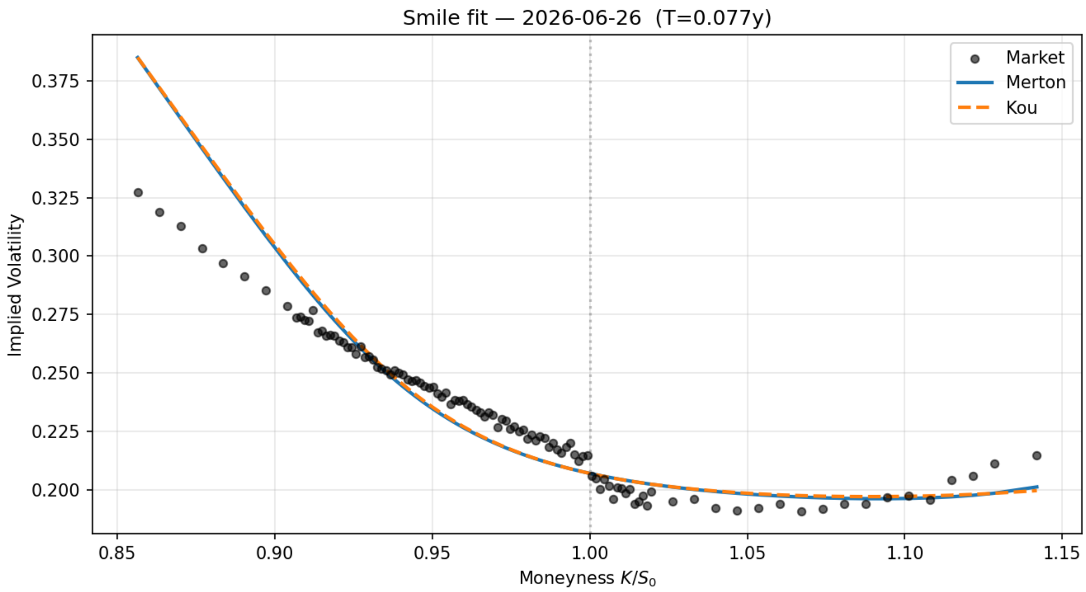

Figure 1: Market vs Merton and Kou implied volatilities for the one-month expiry. Black dots: market quotes; solid line: Merton fit; dashed line: Kou fit.

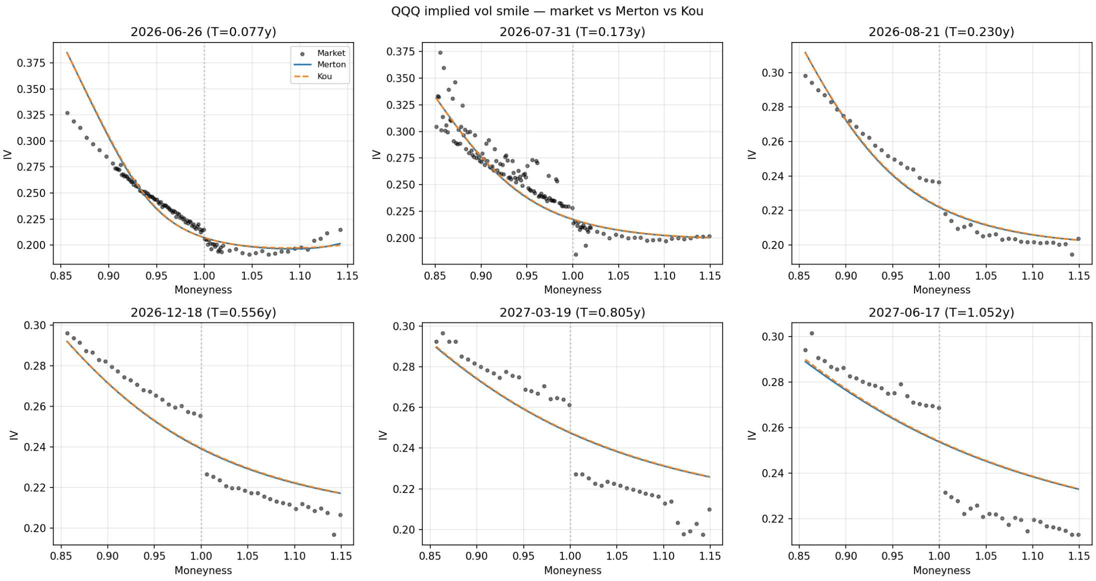

Figure 2: Smile fit across all six calibrated maturities.

The Identification Problem

The project specification asks us to demonstrate that σ and λ are not jointly identified from option prices. Our calibration produces a stronger result: the more severe non-identification is between λ and μ_J, evidenced by the boundary behaviour above and confirmed by three independent diagnostics.

Diagnostic 1: Boundary Attachment

The unregularised optimum lies at $\mu_J = -0.4954$, i.e., effectively on the lower bound $\mu_J = -0.50$. We did not impose this bound *a priori* to constrain the problem; it was the widest economically plausible range. Nevertheless, the optimiser is unwilling to move away from it, indicating that the data prefers a corner solution.

Diagnostic 2: Two-Dimensional Loss Surface

Fixing (σ, σ_J) at their calibrated values and sweeping an 18×18 grid in (λ, μ_J) on the sub-sampled dataset reveals a curved low-loss valley running from the bottom-left to the upper-right of the plane. Within RMSE $+0.002$ of the grid minimum we find 7 distinct parameter pairs, all producing economically indistinguishable fits. The market data does not contain enough information to discriminate among them.

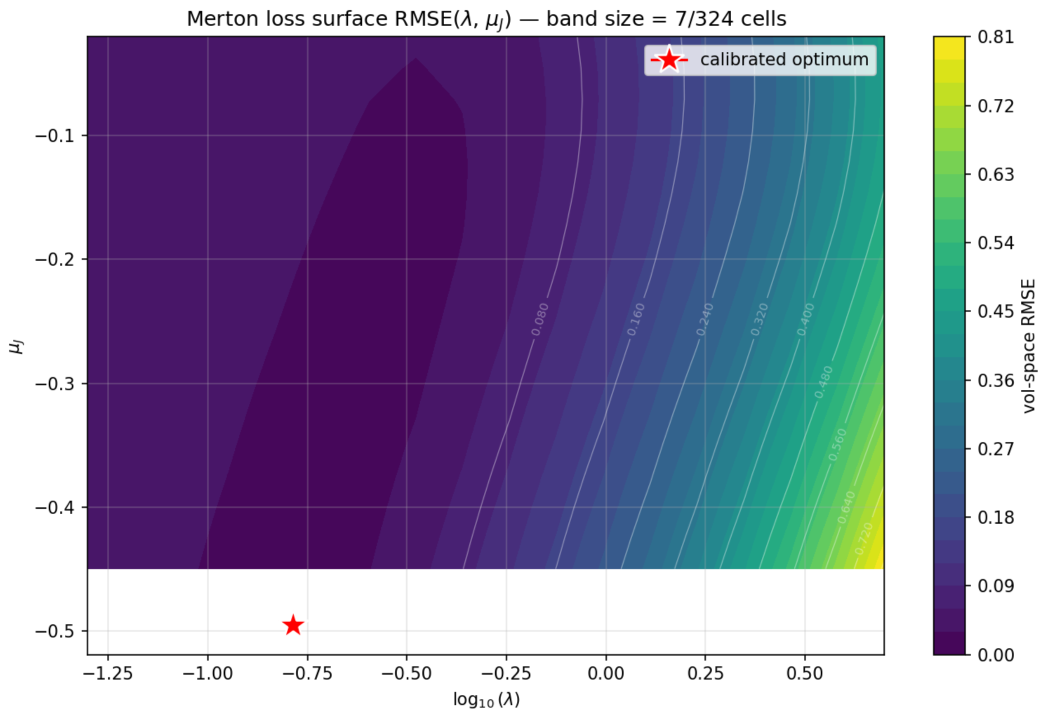

Figure 3: Merton vol-space RMSE surface in $(\log_{10}\lambda, \mu_J)$ with (σ, σ_J) fixed at calibrated values. White contours show iso-loss lines; the red star marks the calibrated optimum.

Diagnostic 3: L1-Regularisation Path

Adding a penalty $\alpha\,|\mu_J|$ to the loss pulls the optimum away from the boundary, as expected for an ill-posed inverse problem. The regularisation path is computed on the stratified 96-quote sub-sample used for the global search to keep its five calibrations tractable in wall-clock time; the $\alpha = 0$ row is therefore numerically close to, but not exactly identical to, the full-surface main optimum in Table above. The qualitative trend (boundary attachment and recovery under penalty) is unaffected.

Table 2: Regularisation path. The L1 penalty $\alpha\,|\mu_J|$ added to the loss pushes the optimum out of the boundary; as α grows the optimum approaches the literature-consistent regime at a modest cost in fit.

α	σ	λ	μ_J	σ_J	RMSE (unreg.)
0.000	0.1863	0.163	−0.5000	0.3051	0.01398
0.005	0.1794	0.328	−0.2718	0.2493	0.01448
0.010	0.1741	0.486	−0.1967	0.2161	0.01502
0.020	0.1659	0.803	−0.1329	0.1791	0.01593
0.050	0.1369	2.502	−0.0606	0.1164	0.01851

At $\alpha = 0.05$ the solution lands at $\mu_J \approx -0.06$ and $\lambda \approx 2.5$, the literature-consistent regime of "moderate jumps several times per year" documented in Andersen, Benzoni and Lund (2002) and Eraker, Johannes and Polson (2003). The cost is a 32% increase in RMSE ($0.01400 \to 0.01851$), which we view as a favourable trade-off: a modest deterioration in fit buys parameters that are both economically plausible and statistically stable.

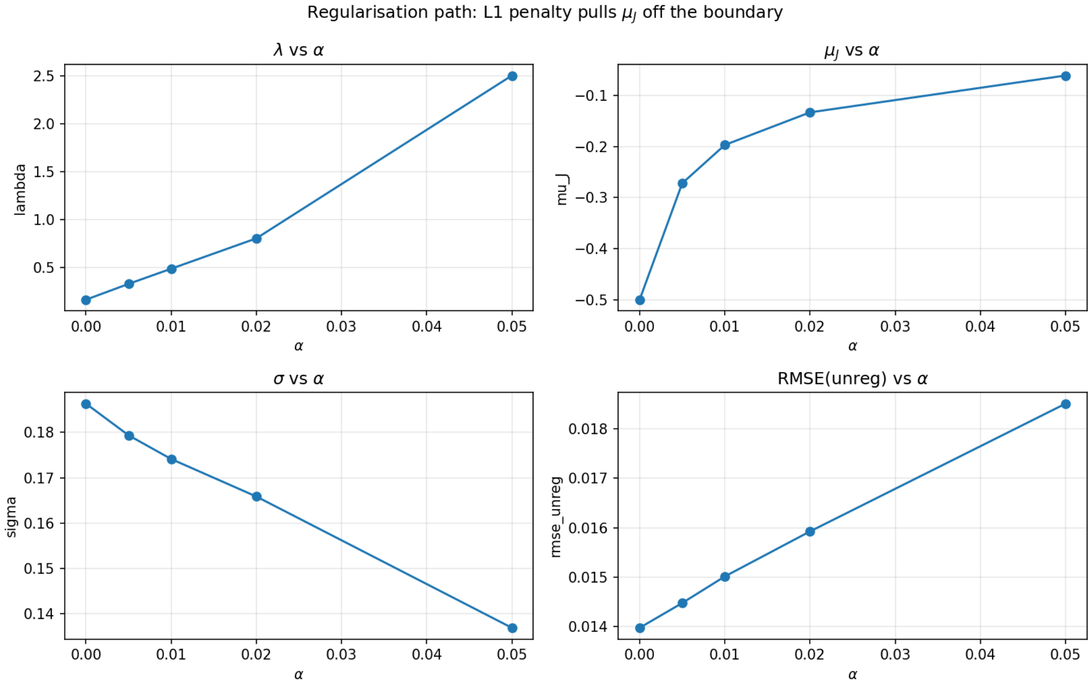

Figure 4: Regularisation path. Each panel shows one Merton parameter as a function of the L1 penalty strength α. Bottom-right: the unregularised RMSE deteriorates only modestly as α grows.

The economic content of this exercise is the *peso problem* of Backus, Chernov and Martin (2011): the smile of an equity index can be generated either by frequent, mild jumps or by rare, catastrophic ones, and the data alone cannot tell the two regimes apart.

Kou Comparison

We re-calibrated the same QQQ surface to the Kou (2002) double-exponential jump model, which replaces Merton's normal log-jump with the asymmetric density

$$f_J(j) \;=\; p\,\eta_1 e^{-\eta_1 j}\,\mathbf{1}_{j\geq 0} + q\,\eta_2 e^{\eta_2 j}\,\mathbf{1}_{j<0}, \qquad p+q=1,\;\eta_1>1,\;\eta_2>0.$$

The characteristic function from W2 substitutes directly into the COS kernel; calibration uses the same two-stage protocol.

Table 3: Calibrated Kou parameters (QQQ, 2026-05-29). The "$\approx 41\%$" figure for $1/\eta_2$ refers to the mean *log*-jump magnitude on the downside; the corresponding expected percentage price drop conditional on a downward jump is $1-\eta_2/(\eta_2+1)\approx 29\%$.

Parameter	Value	Interpretation
σ	0.1813	diffusion volatility
λ	1.825	jump intensity (jumps per year)
p	0.875	probability of upward jump ($q = 0.125$)
η_1	46.17	upward tail; mean upward log-jump $1/\eta_1 \approx 2.2\%$
η_2	2.44	downward tail; mean downward log-jump magnitude $1/\eta_2 \approx 41\%$
RMSE	0.01398	vol-space RMSE on 424 quotes

The economic structure is striking: 87.5% of jumps are small upward moves (mean log-jump

$\approx 2.2\%$), but the remaining 12.5% are large downward moves whose mean log-magnitude is about 41% — which corresponds to an expected percentage drop, conditional on a down-jump, of $1 - \eta_2/(\eta_2 + 1) \approx 29\%$. The Kou compensator is

$$\kappa^Q = \frac{p\,\eta_1}{\eta_1 - 1} + \frac{q\,\eta_2}{\eta_2 + 1} - 1 = \frac{0.875 \cdot 46.17}{45.17} + \frac{0.125 \cdot 2.44}{3.44} - 1 \approx -1.7\%.$$

The mean percentage jump is small in magnitude because the frequent small upward jumps and the rare large downward jumps approximately offset each other in mean. The variance contribution, by contrast, is dominated by the heavy left tail.

Goodness-of-Fit Comparison

The table below places Merton and Kou side by side. Kou's vol-space RMSE is marginally lower (0.01398 vs 0.01400), but the Akaike information criterion penalises Kou for its additional parameter and favours Merton (AIC $= -3612.02$ vs -3610.96).

Table 4: Model comparison. AIC penalises Kou for the extra parameter despite a marginally better in-sample fit.

Model	k params	RMSE	AIC
Merton	4	0.01400	-3612.02
Kou	5	0.01398	-3610.96

This result is not the textbook expectation that Kou should dominate Merton on equity surfaces; instead it reflects identification on the model-class level. Merton's $\mu_J \approx -0.50$, $\lambda \approx 0.16$ optimum (rare large jumps) and Kou's $1/\eta_2 \approx 0.41$, $\lambda \approx 1.83$ optimum (occasional very large downward jumps) are two different mathematical languages describing the same underlying realisations. The QQQ surface, taken on a single day, does not contain enough information to discriminate the two regimes.

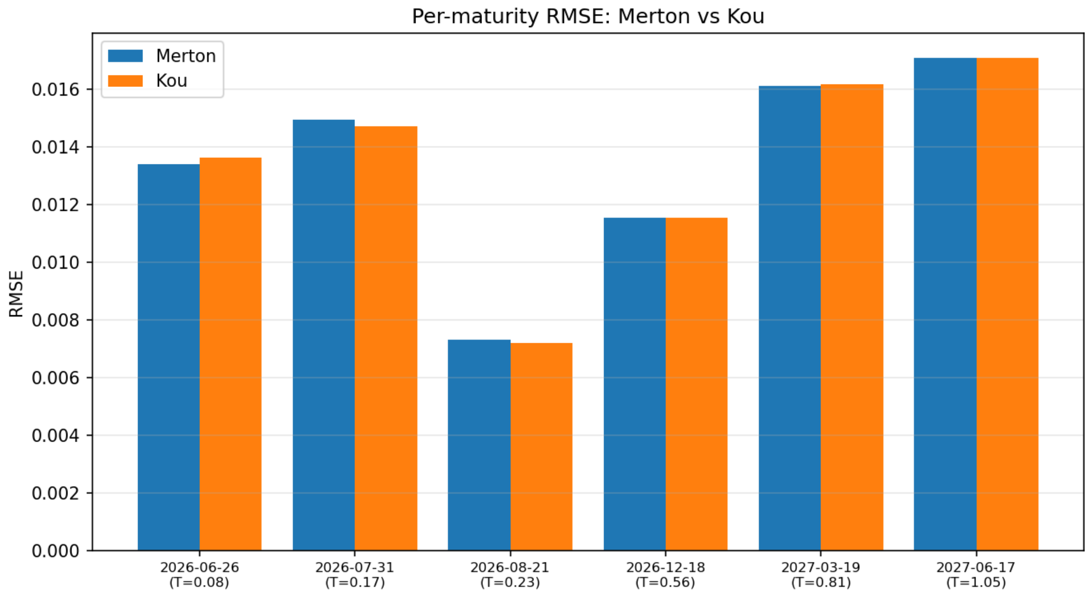

Figure 5: Per-maturity RMSE: Merton (blue) vs Kou (orange).

Jump Risk Premium

The final element of the calibration compares the risk-neutral jump parameters from the option calibration with the physical parameters from the W5 jump-detection pipeline. The W5 BNS-style diagnostic produces two jump-day classifications: a *liberal* flag (Z-score above 2.0, 99 jump days) and a *conservative* flag (Z-score above 2.0 *and* return above 3× rolling-vol, only 4 jump days). We report the premium against both, since neither corresponds exactly to the events the risk-neutral λ^Q is pricing.

Intensity

The risk-neutral intensity is $\lambda^Q = 0.163$ jumps per year. The liberal physical intensity is $\lambda^{P,\text{lib}} = 34.08$ and the conservative one $\lambda^{P,\text{cons}} = 1.38$. A direct difference $\lambda^Q - \lambda^P$ is meaningless because the three quantities count different events: liberal-P counts any volatility-test rejection (average $|r| \approx 1.3\%$), conservative-P counts only the very largest moves (average $|r| \approx 6.3\%$), while λ^Q measures the intensity of much rarer events ($|\kappa^Q| \approx 36\%$) required to bend the IV smile.

Jump-Variance Risk Premium

The cleaner, units-free metric is the ratio of total annualised jump variance under each measure:

$$\text{JVRP} \;=\; \frac{\lambda^Q \cdot \mathbb{E}^Q[J^2]}{\lambda^P \cdot \mathbb{E}^P[J^2]}.$$

Plugging in $\mathbb{E}^Q[J^2] = \mu_J^2 + \sigma_J^2 = 0.357$, and the empirical second moments $\mathbb{E}^{P,\text{lib}}[J^2] = 3.7 \times 10^{-4}$, $\mathbb{E}^{P,\text{cons}}[J^2] = 3.96 \times 10^{-3}$:

$$\text{JVRP}_{\text{lib}} \;=\; \frac{0.163 \times 0.357}{34.08 \times 3.7 \times 10^{-4}} \;=\; \frac{0.0583}{0.0127} \;=\; 4.57,$$

$$\text{JVRP}_{\text{cons}} \;=\; \frac{0.163 \times 0.357}{1.38 \times 3.96 \times 10^{-3}} \;=\; \frac{0.0583}{0.0055} \;=\; 10.69.$$

Investors price jump variance at between 4.6 and 10.7 times its historical realisation, depending on how broadly "a jump" is defined. Both numbers are large positive premia and consistent in sign with the empirical estimates of Pan (2002), Eraker (2004), and Bollerslev and Todorov (2011) for the S&P 500, although our magnitudes are at the upper end of the published range, a direct consequence of the peso-style calibration with very rare but very large risk-neutral jumps.

This finding directly underwrites the W8 alpha strategy: short-volatility positions in QQQ index options harvest, on average, the jump variance premium documented here, while bearing the corresponding jump risk in the left tail of their P&L distribution.

Limitations and W7 Hand-Off

Our calibration has five limitations worth flagging for the volatility analysis in W7 and the hedging study in W8.

1. **American-style options approximated as European.** QQQ ETF options are American. We use European COS pricing because the calibration restricts to out-of-the-money quotes, where the early-exercise premium is small. A production implementation should add an American-style correction (Bjerksund–Stensland) or work on forwards.

2. **No dividend yield.** The QQQ ETF pays a small quarterly dividend (˜0.6% annualised) that we do not subtract from the discount rate. This biases the implied σ marginally upward for longer-dated options. The proper fix is to calibrate on the forward $F = S_0 e^{(r-q)T}$ rather than the spot.

3. **Boundary behaviour and identification.** The Merton optimum lies on the lower bound for μ_J and would walk further if allowed. All downstream Greeks, hedging ratios, and P&L distributions should be reported at *both* the unregularised parameters and at the regularised parameters of the previous subsection, to avoid relying on a single corner solution.

4. **Short-maturity smile.** Both Merton and Kou fit the shortest maturity systematically worse than longer-dated options. W7's smile-decay analysis should document this and point to richer models (Bates, CGMY) that address it.

5. **Single-day snapshot outside trading hours.** Only 2% of our quotes use a simultaneous live mid; the remainder fall back to last-trade prices. A multi-date study during NYSE hours would tighten all conclusions, particularly the Kou-vs-Merton comparison.

The calibrated parameters are exposed to W7 through `src/iv_surface.py`, which provides:

- `get_merton_params()` and `get_kou_params()`, returning the calibrated dictionaries;
- `get_merton_params_regularised(alpha)`, returning the regularised alternatives at any α from the path above;
- `merton_iv(...)`, `kou_iv(...)`, returning the model implied volatility of a single option;
- `merton_iv_surface(S0, r, K_grid, T_grid, params)`, returning a two-dimensional model IV grid suitable for direct comparison with the market surface in W7.

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